MLFF-DATA9B2 specification: execution, evaluation, aggregation, committee, and freeze

Version: 0.20.57a0

Purpose

DATA9B2 closes the production-training control loop after DATA9B1. It does not make a model scientifically acceptable by itself. It makes every training run, checkpoint evaluation, protocol comparison, committee member, and protocol freeze an immutable, lineage-checked record.

Inputs

DATA9B2 consumes only:

- a passed `ProductionCorpusQualificationRecord`;
- a frozen `TrainingCampaignPlan` and exact `TrainingCampaignRunPlan` objects;
- exact DATA8 `MaceJobArtifact` files;
- the precision-aware `mdstats-mace-train`, `mdstats-mace-eval`, and `mdstats-mace-select-head` wrappers;
- checkpoint catalogs and monitor artifacts whose digests match the campaign;
- the frozen `CheckpointMetricPolicy` and replay-retention identity.

Supervised execution

`TrainingExecutionPolicy` freezes the wrapper name, retry limit, timeout, restart behavior, checkpoint glob, and environment evidence allowlist. Each attempt is a `TrainingRunAttemptRecord`; the complete supervised result is a `TrainingRunExecutionRecord`.

`execute_training_run` shall:

1. verify the DATA8 config bytes against the job artifact;
2. launch only the declared mdstats wrapper;
3. record exact argv, working directory, environment digest, timestamps, elapsed time, return code, stdout, and stderr for every attempt;
4. persist a fail-closed execution record after every failed or timed-out attempt, so an interrupted supervisor can resume from immutable attempt evidence;
5. terminate the complete POSIX process group on timeout, first with `SIGTERM` and then with `SIGKILL` after the frozen grace period when necessary;
6. add `--restart_latest` only on policy-authorized retries;
7. inventory successful checkpoint bytes by SHA-256;
8. reject a successful process that produced no candidate checkpoint when the policy requires checkpoints;
9. reverify all checkpoint bytes before reusing a completed execution record.

A failed or timed-out process is evidence, not an exception that can be silently ignored.

Automatic checkpoint evaluation

`CheckpointEvaluationPolicy` freezes the inputs used to create a `CheckpointEvaluationRecord`:

- target and replay head names;
- reference energy, force, and stress keys;
- focus atomic numbers;
- condition-group keys;
- replay metric and baseline floor;
- combined-metric weights;
- device and dtype.

`evaluate_mace_checkpoint` shall evaluate the exact checkpoint bytes with the critical-FP64 patch installed. It materializes:

- energy MAE per atom;
- force-component RMSE;
- focus-species force RMSE;
- stress RMSE when labels exist;
- worst declared-condition force RMSE;
- target combined loss;
- replay baseline and candidate metrics;
- replay degradation fraction;
- monitor, model, and policy hashes.

For pseudo-label replay, the baseline floor prevents division by a numerically zero foundation-model error from producing an undefined ratio. Absolute baseline and candidate metrics remain serialized.

Fold and seed aggregation

`ProtocolVariantAggregate` groups one seed's protocol-matched final run and all cross-validation folds. It records fold values, mean, standard deviation, and worst fold.

`ProtocolFamilyAggregate` combines seed variants only when training mode, selection size, protocol family, campaign, and primary metric all match.

`LearningCurveRecord` orders comparable protocol families by selection size. It never assumes monotonic improvement.

`ProtocolComparisonRecord` ranks complete family aggregates by the frozen mean cross-validated primary metric, then worst-seed metric, then content digest. Naive and replay campaigns are different protocol families.

Committee export

`CommitteeExportPolicy` requires the precision-aware `mdstats-mace-select-head` wrapper and freezes the target head and export device.

Each `CommitteeMemberRecord` binds:

- final run and seed;
- checkpoint-selection record;
- source checkpoint SHA-256;
- extracted target-head SHA-256 and size.

`CommitteeIdentity` requires exact campaign seed coverage and distinct exported model bytes. Cross-validation checkpoints cannot become committee members.

Protocol freeze

`ProtocolFreezeRecord` owns only frozen identities:

- passed production qualification;
- campaign plan;
- source, frame, and DATA5 lineages;
- selected protocol comparison and family aggregate;
- committee identity and member model hashes;
- exact final checkpoint-selection records.

It shall not contain locked-test results.

`EvaluationActivationDecision` may activate sealed evaluation only when the freeze, committee, frame catalog, DATA5 bundle, and activation requirement all match. Otherwise it records explicit rejection reasons.

Fail-closed requirements

DATA9B2 shall reject:

- raw `mace_run_train` or unqualified head-selection wrappers;
- changed DATA8 config bytes;
- missing checkpoint output after successful training;
- checkpoint/model hash mismatch;
- missing target or replay monitor evidence;
- incomplete fold/final or seed coverage;
- mixed protocol identities in an aggregate;
- committee members from fold jobs;
- duplicate committee seeds or exported model bytes;
- protocol freeze before full DATA9A passage;
- locked evaluation whose frame or DATA5 lineage differs from the freeze.

Scope boundary

This stage implements the production execution and freeze machinery and tests it with bounded external processes plus the supplied real MACE multi-head smoke model. It does not claim that the 2,734-frame production DATA6-DATA8 realization or a long multi-seed production campaign has been executed. Scientific acceptance, calibration, long MD validation, and active learning remain DATA10 and DATA11 responsibilities.